



RF-LAB
VECTOR NETWORK ANALYZER
EXPERIMENTAL MANUAL

...A MESSAGE FROM

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We strive to be the best in what we do and maintain high standards in the areas of

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As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting Cords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting Cords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions' sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions' judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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Technical Specifications for VNA

Specifications

- Frequency range 1-3000 MHz
- Calibration using open-short-load for accurate results
- Range of Z from 1 to 1000 ohm
- Two ports VNA with S11 and S21; displayed and save results
- Low power consumption, 370mA @ 5 V
- SMA connectors for better isolation
- Dynamic range up to 70 dB @ 500 MHz
- User friendly interface for PC: Windows
- Integrated Smith chart in software
- Export data in several formats – JPEG, EXCEL, ZPLOT, S2P, PDF

EXPERIMENT NO.1

EXPERIMENT NO: 1

NAME

Calibration of Vector Network Analyzer(VNA).

AIM:

The object of this experiment is to introduce the students to familiarize calibration of vector network analyzer (VNA).

OBJECTIVES:

- To understand the purpose of a calibration kit.
- Basic knowledge on Vector network analyzer measurement and concept.
- Verification of the calibration of vector network analyzer.

EQUIPMENTS

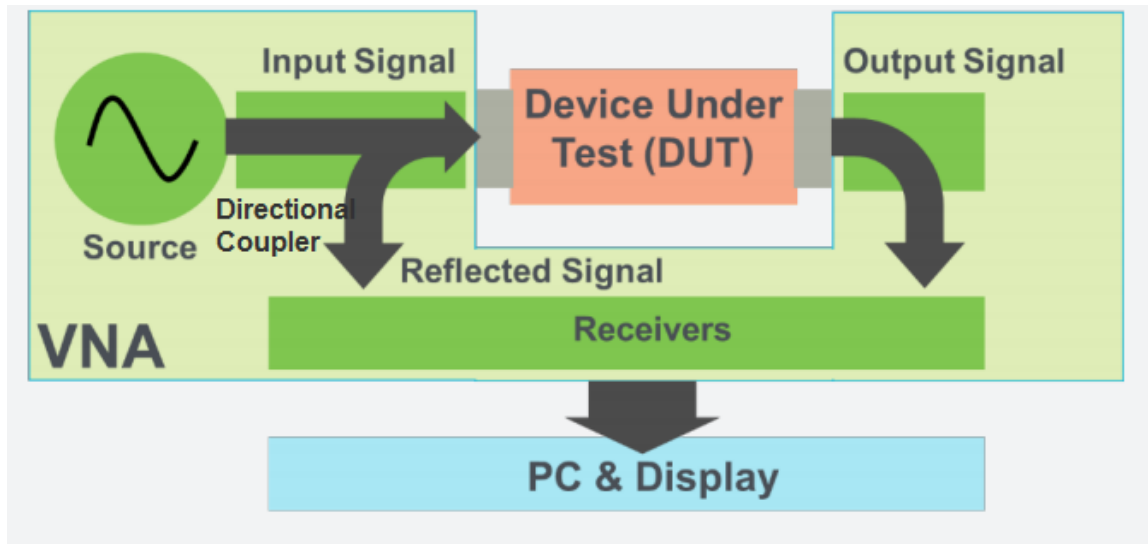
(a) List of Equipments /Accessories needed for performing the experiments

DUT, Equipments and Accessories	Quantity
Vector Network Analyzer (VNA)	1
Calibration Kits (OPEN,SHORT,LOAD and THROUGH)	1
RG 316 SMA Male to SMA Male 50 ohm coaxial cable	2

1) VNA Introduction

The vector network analyser Product utilises the concept of measuring the transmitted and reflected waves as a signal passes through a device under test.

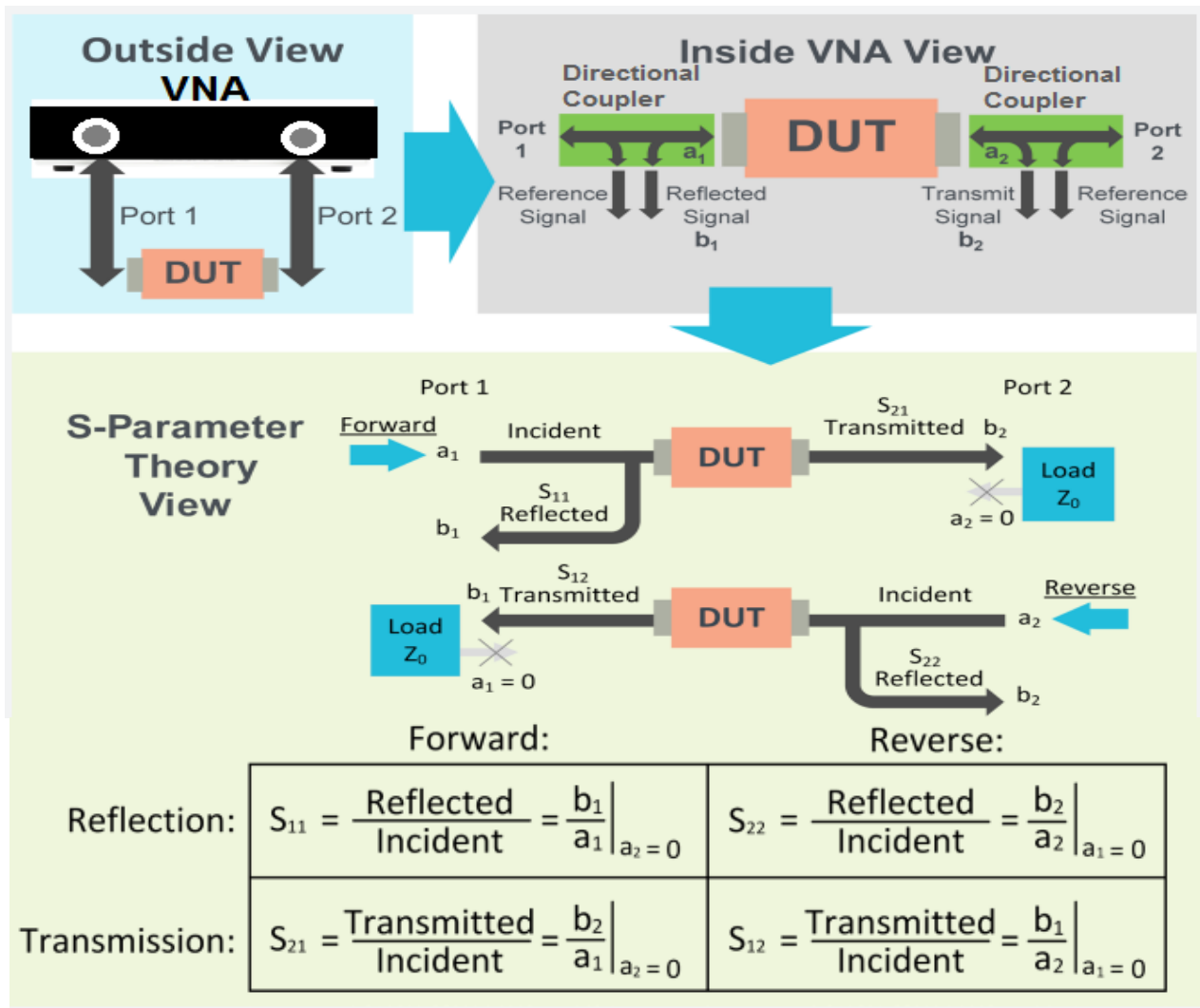
VNAs are used in a wide range of RF and high frequency design applications. VNA are used to test component specifications and verify design simulations to make sure systems and their components work properly together. The commonly VNAs are used for performance testing of components such as amplifiers, filters, antennas, cables, mixers, etc. In some cases, VNA are even used in field operations to verify and troubleshoot deployed RF and microwave systems. Vector Network Analyser can be measured as Return Loss, VSWR, Impedance, Reflection coefficient and phase etc.



A Vector Network Analyzer contains both a source and receiver, used to determine changes to signal caused by the device-under-test or DUT.

Vector Network Analyzer (VNA) perform two types of measurements – transmission and reflection.

There are a variety of different VNAs available on the market, each with a different number of ports for which the signal flows. For 1-port VNA, the DUT is connected to the input side and only the reflected signals can be measured. For 2-port VNA, the DUT can be connected to either port in either direction because the instrument has the capability of reversing the signal flow so that the reflections at both ports (S_{11} and S_{22}), as well as the forward and reverse transmissions (S_{21} and S_{12}), can be measured.



In the Forward case, Port 1 is transmitting the a_1 signal and a matched load is applied to Port 2, resulting in zero signal reflection at the load ($a_2 = 0$). S_{11} corresponds to the reflection coefficient at Port 1, or ratio of b_1 over a_1 . S_{21} is the forward transmission coefficient through the DUT and is the ratio of b_2 over a_1 .

In the Reverse case, Port 2 is transmitting the a_2 signal and a matched load is applied to Port 1 ($a_1 = 0$). S_{22} corresponds to the reflection coefficient at Port 2, or ratio of b_2 over a_2 . S_{12} is the reverse transmission coefficient through the DUT and is the ratio of b_1 over a_2 .

2) Why Is Calibration Needed?

Components of the measurement setup such as imperfect connectors, cabling, and even the response of the test instruments can introduce errors into measurements. For both transmission and reflection measurements, impedance mismatches within the test setup cause measurement uncertainties that appear as ripples superimposed on the measured data. These errors can distort the signal and make it difficult to determine which reflections are from the DUT and which are from other sources.

Calibration is required for accurate measurements. Even though calibration does take a few minutes to complete, it saves time and money compared to costs associated with erroneous measurement data.

3) VNA Calibration Procedure

The calibration processes for the S-parameter characterization of the passive circuits are described below. These are only described as general information, the actual process described in the User Manual of the Instrument at the user end should be followed.

In some VNA's only one port (namely Port 1) is capable of applying a RF signal while the other port acts as a receiving port. The two port calibration process for the above described VNA is called **One Path Two Port Calibration**.

One-Path Two-Port Calibration

One-path two-port calibration is used for measurements of the DUT parameters in one direction, e.g. S_{11} and S_{21} . This method involves connection of the three calibration standards to port 1, and connection of a THROUGH standard between port 1 and port 2 (See figure a).

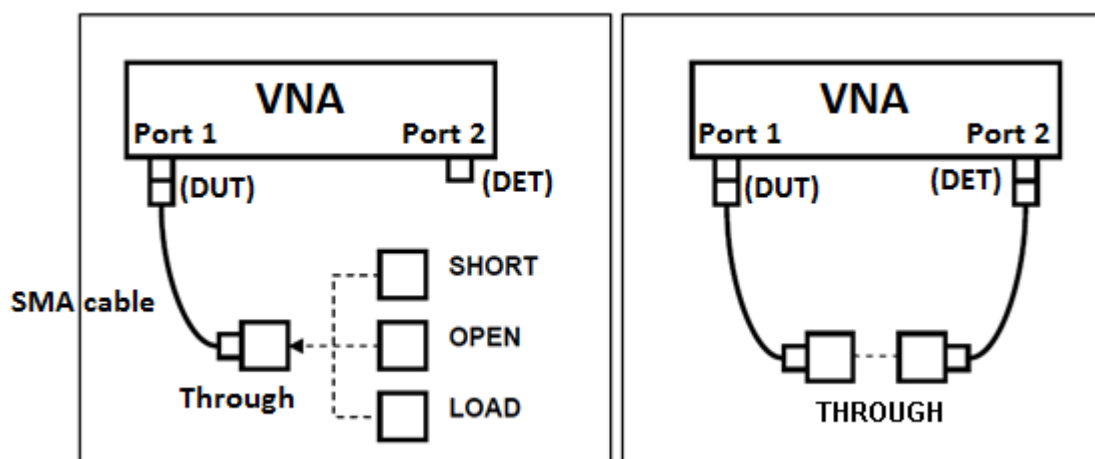
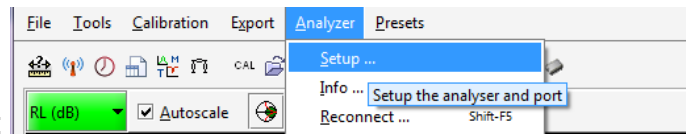


Fig 1 : One-Path Two-Port Calibration

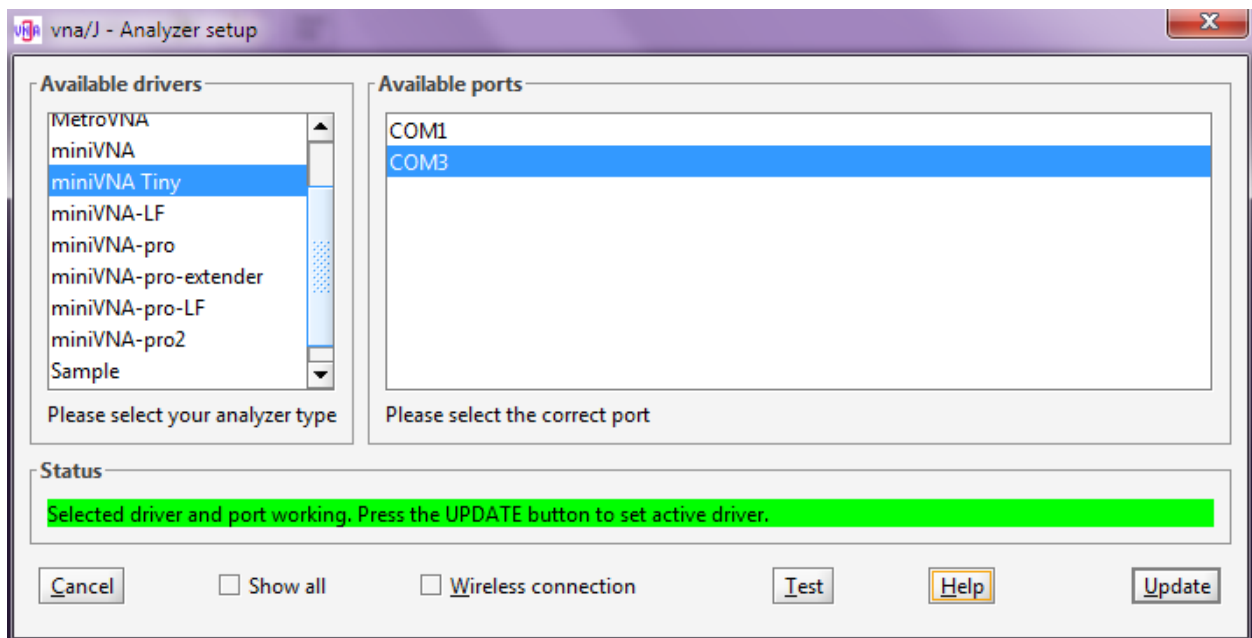
Experimental Setup & Procedure Using VNA

1. Plug-in the VNA into a free USB port on your PC.
2. Start the application **vnaJ.3.2.10**

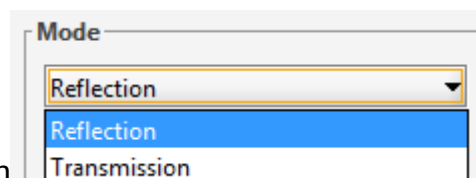
3. Select the driver configuration:



4. Select your analyser type and the used communication port and Select the TEST button to confirm the selection



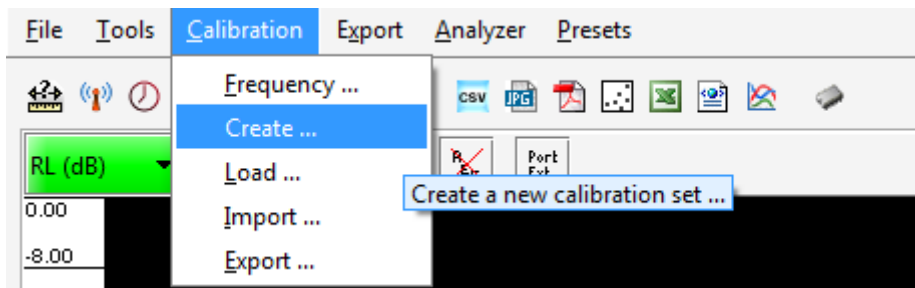
5. If test was run successful simply select UPDATE.



6. Select the mode Reflection

7. The next step is to calibrate the VNA. This calibration technique permits the accurate measurement of the device characteristics and compensates for the effects of the cable and connector properties on the measurement.

Then go to the calibration menu and create a calibration for **reflection** measurement.



You have 3 steps that may be active

- **Read OPEN:** open (nothing connected on port1(DUT) and port 2 (DET))
- **Read SHORT:** 0 Ohm where Short is connected on DUT
- **Read LOAD:** 50 Ohm where the unnamed terminator is connected on DUT

DUT Port 1 means ("device under test") and DET Port 2 means ("detector")

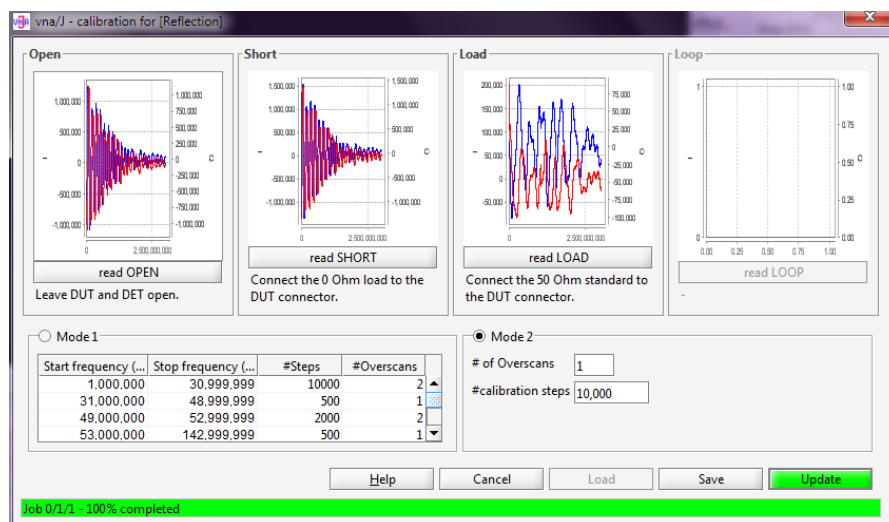
8. Follow the instructions for the selected mode

9. With the coaxial SMA cables connected to the port1(DUT) and port 2 (DET) ports on the VNA and the free ends unconnected or connect **OPEN (O)** to DUT, click on the "read OPEN" button.

10. Connect **Short(S)** is called 0 Ohm on Port 1 (DUT)

11. Connect **Load(L)** is called 50 Ohm on Port 1 (DUT)

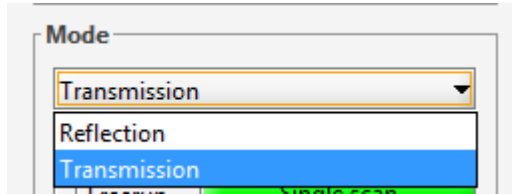
After this step, the calibration window will resemble this one.



12. To Save the calibration, click Save. Choose an appropriate file name for the calibration data, enter an optional comment and click the "Save" button to return to the calibration window. You may reload this data file in the future for measurements with these same

cables. Click the "Update" button to load the new calibration data into the VNA and to return to the main vna/J window.

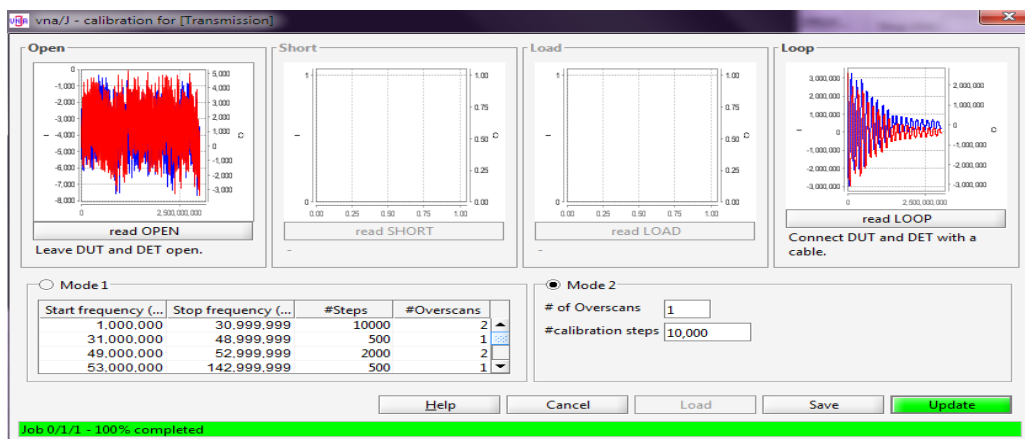
13. Select "Transmission" in the Mode drop-down box.



14. Go to Calibration Menu and create calibration for **transmission** measurement.

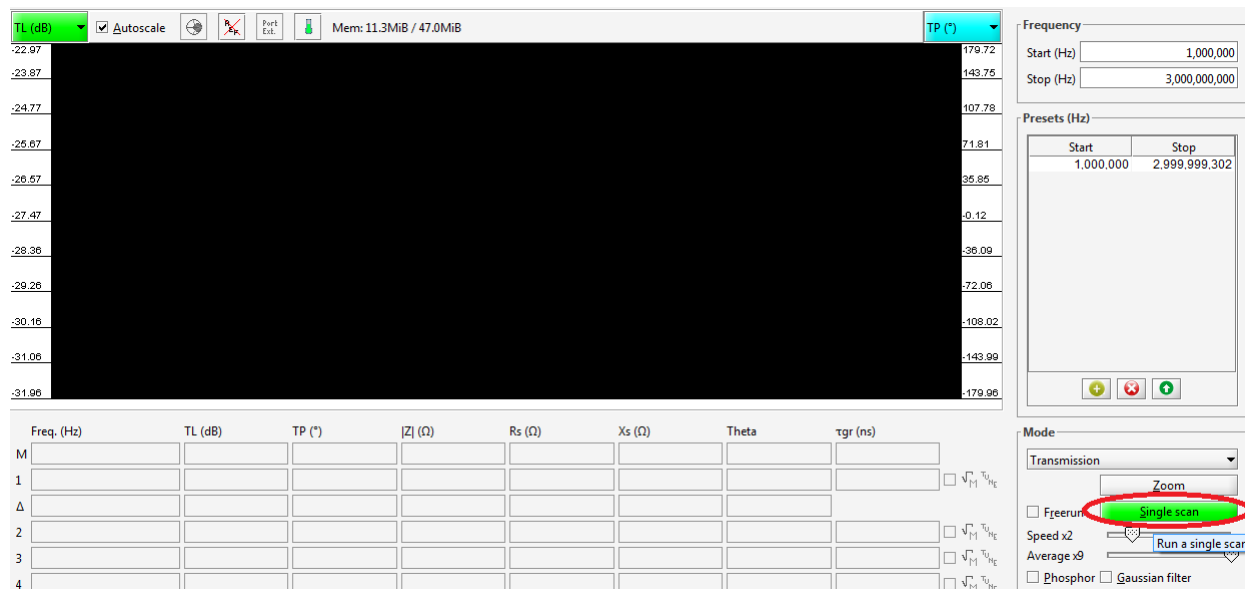
15. Connect **OPEN(O)** on Port 1 (DUT)

16. Connect DUT (Port-1) and DET (Port-2) with through a piece of RG 316 SMA cable as short as possible, the calibration window will resemble this one.

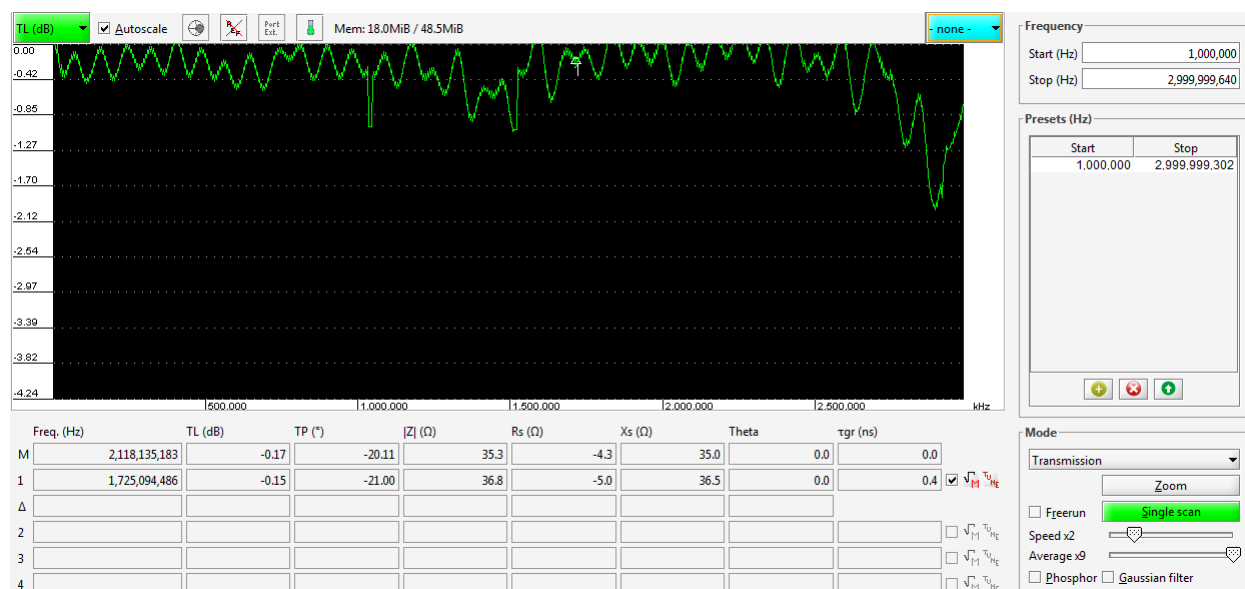


17. To Save the calibration, click Save. Choose an appropriate file name for the calibration data, enter an optional comment and click the "Save" button to return to the calibration window. You may reload this data file in the future for measurements with these same cables. Click the "Update" button to load the new calibration data into the MiniVNA and to return to the main vna/J window.

18. Click on single scan to run transmission (S21) of Coaxial cable. Click to to freerun until the graph steady and then untick if graph is stable.



19. After the scan, the graph area displays a transmission loss of -0.38 dB and phase shift of 1.5° over the frequency range from 1MHz-3GHz. This validates the accuracy of the calibration.



20. export the measurement calibration data in several formats JPEG, EXCEL, ZPLOT, S2P, PDF.

OBSERVATION TABLE

Frequency (F) (MHz)	Insertion loss/transmission (S ₂₁)dB
100	
500	
1000	
1500	
2000	
2500	
3000	

Summarize for transmission loss (S₂₁) at different frequency.

EXPERIMENT NO.2

EXPERIMENT NO. 2

NAME

Measurement parameters like Return loss, VSWR, Bandwidth and Impedance of Rectangular Microstrip Antenna(RMSA) at 2.4GHz

OBJECTIVES

- Antenna characteristics like Measurement of Return Loss, VSWR, Phase and bandwidth.
- Study of Basic antenna parameters.

EQUIPMENTS

List of equipment's needed for performing the experiments

DUT, Equipment's and Accessories	Model Name	Quantity
Planer Antenna	RMSA	1
Vector Network Analyzer	VNA	1
SMA Coaxial Cable	Standard 50 ohm RG 316 SMA Male To SMA male	1

ANTENNA BASIC PARAMETERS

1. Return Loss

This bounce back reflection is called “return”. Return loss is the measure of how small the “return” or reflection is. We want a small return, so a large loss on the return “echo” is good. Smaller return loss is bad, and means less energy is going into our antenna. RF engineers often measure return loss on a “dB” logarithmic scale, which can make it seem more complicated than it really is. However, just remember better return loss is indicated by bigger return loss numbers, and that is better for your antenna. Here are some examples of the logarithmic scale, or loss in decibels:

Return Loss in dB	What It Means	VSWR Number
0 dB	100% reflection, no power into the antenna, all reflected back	Infinite
-1 dB	80% reflection, 20% power into the antenna	17
-2 dB	63% reflection, 37% power into the antenna	9
-3 dB	50% reflection, 50% power into the antenna	6
-5 dB	32% reflection, 68% power into the antenna	3.5
-6 dB	25% reflection, 75% power into the antenna	3
-8 dB	16% reflection, 84% power into the antenna	2.3

-10 dB	10 dB (10% reflection, 90% power into the antenna)	2
-15 dB	15 dB (3% reflection, 97% power into the antenna)	1.4
-20 dB	20 dB (1% reflection, 99% power into the antenna)	1.2

2. VSWR

Voltage Standing Wave Ratio (VSWR) is an indication of the quality of the impedance match. VSWR is often abbreviated as SWR. A high VSWR is an indication the signal is reflected prior to being radiated by the antenna. VSWR and reflected power are different ways of measuring and expressing the same thing.

A VSWR of 2.0:1 or less is often considered acceptable. Most commercial antennas are specified to be 1.5:1 or less over some bandwidth. Based on a 100 watt radio, a 1.5:1 VSWR equates to a forward power of 96 watts and a reflected power of 4 watts, or the reflected power is 4.2% of the forward power.

3. Impedance Matching

Impedance Matching is the process of removing mismatch loss. That is, we want to minimize the reflection coefficient, to reduce the power reflected from the load (the antenna), and maximize the power delivered to the antenna. This is one of the fundamental tasks in getting an antenna to radiate, and hence is one of the more important topics in antenna theory.

To achieve perfect matching, we want the antenna or load impedance to match the transmission line. That is, we want $Z_L = Z_0$ (or $Z_{in} = Z_0$).

Smithchart

Another popular output is for the impedance to be measured on a [Smith Chart](#). A Smith Chart is basically a graphical way of viewing input impedance (or reflection coefficient) that is easy to read. The center of the Smith Chart represented zero reflection coefficient, so

that the antenna is perfectly matched to the VNA. The perimeter of the Smith Chart represents a reflection coefficient with a magnitude of 1 (all power reflected), indicating that the antenna is very poorly matched to the VNA. The magnitude of the reflection coefficient (which should be small for an antenna to receive or transmit properly) depends on how far from the center of the Smith Chart you are. In Smith Chart terms, we want to move the impedance Z_L towards the center of the Smith Chart, where the reflection coefficient Γ is zero.

4. Bandwidth

Bandwidth of an antenna is an important concept. The bandwidth of an antenna refers to the range of frequencies over which the antenna satisfies a particular parameter specification. The parameters generally specified are gain, radiation pattern, the VSWR etc. Most commonly, the VSWR is chosen as the parameter for bandwidth considerations and this bandwidth is called the impedance bandwidth. The lower and upper frequencies conforming to the desired VSWR set the frequency band over which the antenna meets the VSWR specification. A VSWR specification commonly adopted is a 2:1 VSWR, which means that the range of frequencies over which the VSWR is less than 2 is chosen as the bandwidth of operation.

As you can see from the plot, the absolute bandwidth over which the antenna meets the 2:1 VSWR specification is $2.66 \text{ GHz} - 2.38 \text{ GHz} = 0.28 \text{ GHz} = 280 \text{ MHz}$.

Another commonly used way to specify bandwidth is percentage bandwidth, which is explained in more detail in the theory section. It requires specifying the center frequency of operation, which in this case is 2.5 GHz. For this antenna, the percentage bandwidth is:

$$\text{Percentage Bandwidth} = \frac{2.66 \text{ GHz} - 2.38 \text{ GHz}}{2.5 \text{ GHz}} \times 100 = \frac{0.28 \text{ GHz}}{2.5 \text{ GHz}} \times 100 = 11.2\%$$

5.Phase

Phase measurements are made using S-parameters, just like amplitude measurements. A phase measurement is a relative (ratio) measurement and not an absolute measurement. Phase measurements compare the phase of the signal going into a device (the incident signal) to the phase of the device's response signal. The response signal can be either reflected or transmitted. Assuming an accurate calibration has been performed, the difference in phase between the two signals (known as phase shift) is a result of the electrical characteristics of the device under test.

A typical phase measurement required for the tracking of magnitude and phase between several receiver blocks. Phase measurement has the advantages of less variation due to external disturbance, specification of the direction of energy flow.

MEASUREMENT ANTENNA PARAMETERS USING VNA

A. SET UP & PROCEDURE Using Antenna with VNA

Measurement of Return loss, VSWR, Impedance & Bandwidth characteristic of RMSA Antenna.

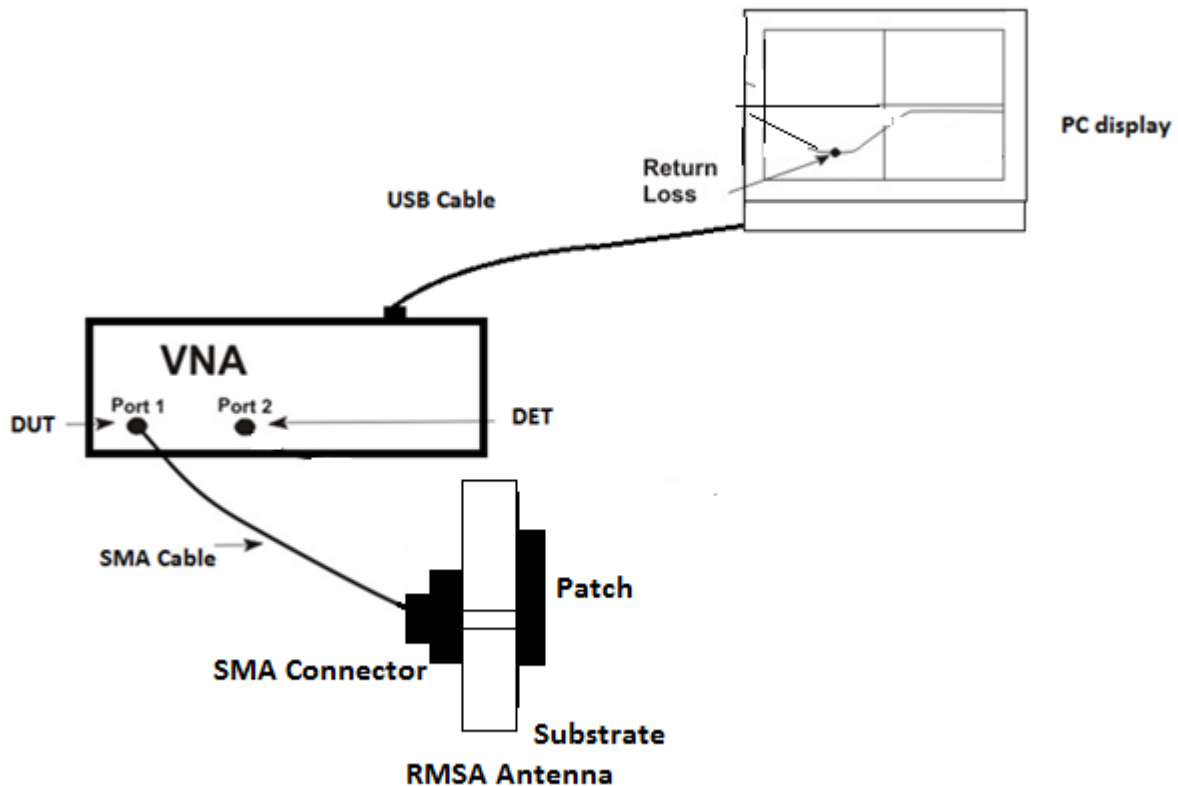


Fig: Setup for Return Loss of RMSA

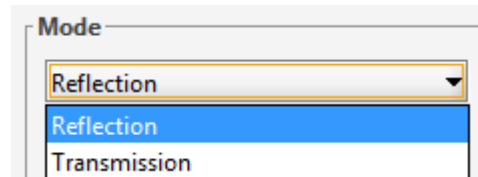
Perform the VNA calibration for the following settings before starting with the experiment and observations. Or you can load existing save calibration file.

- Frequency Range : 2 GHz to 3 GHz
- Calibration Steps : 10,000

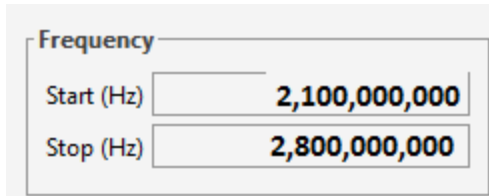
PROCEDURE:

1. Connect Port 1 (DUT) of VNA to Port 1 of the LOW PASS FILTER (i.e. signal from the VNA is applied as input to the DUT) and connect Port 2 of the LPF to Port 2 (DET) of VNA.

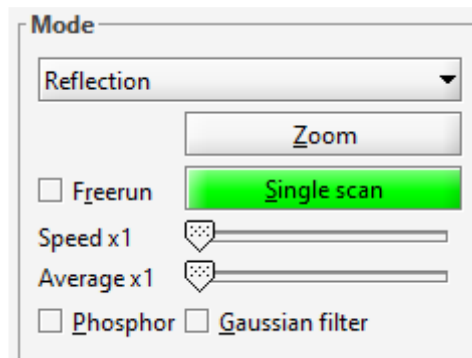
2. Select "Reflection" in the Mode drop-down box.



3. Put Start Freq 2.1GHz and Stop Freq 2.8GHz in right hand side window

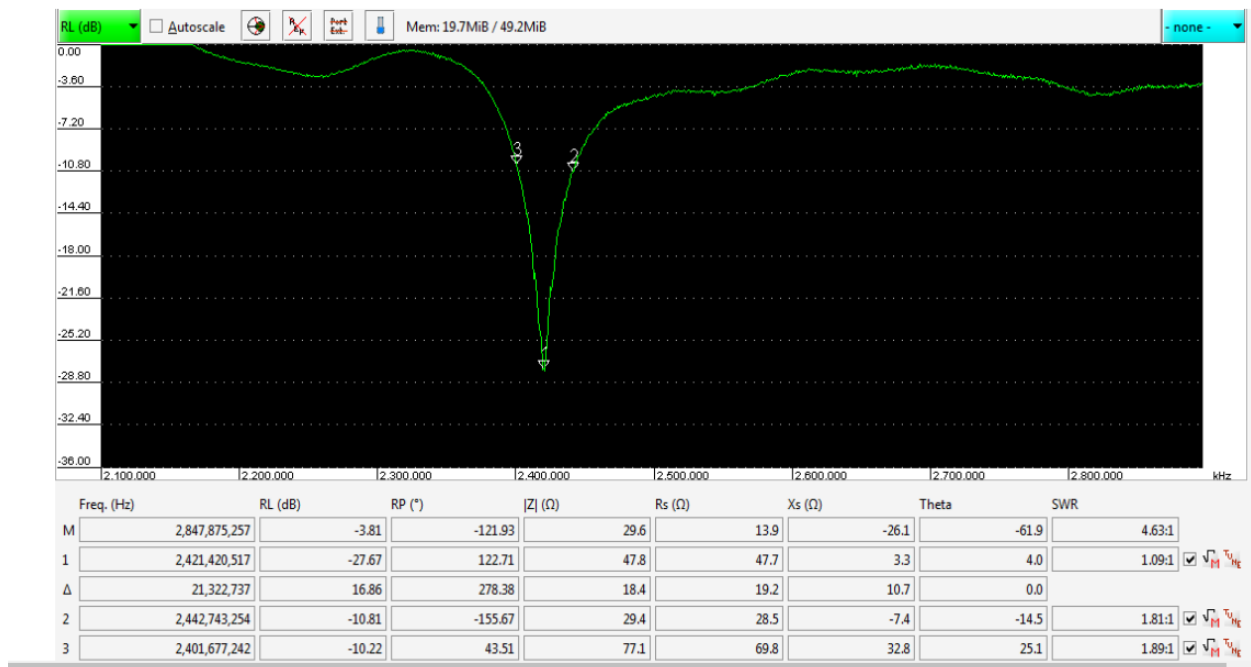


4. Run Single Scan for reflection measurement.



Selecting the checkbox **"Freerun"** enables the continuous measurement with the given parameters in the frequency and mode group. During continuous scanning, most of the menu entries and toolbar buttons are disabled to ensure a correct measurement.

If the checkbox is deselected, the scanning of the VNA stops and the pushbutton for "Single scan" is active



- The display window will show the reflection characteristics or S11 of the RMSA Antenna on a logarithmic scale.
- Insert markers by click on the data plot to study the response of the plot with respect to frequency.

Mouse: Displays values, when the mouse cursor is inside the image panel.

Marker 1: Can be set by moving the mouse into the diagram panel and clicking the **left** mouse but-ton. It can be moved using the mouse-wheel.

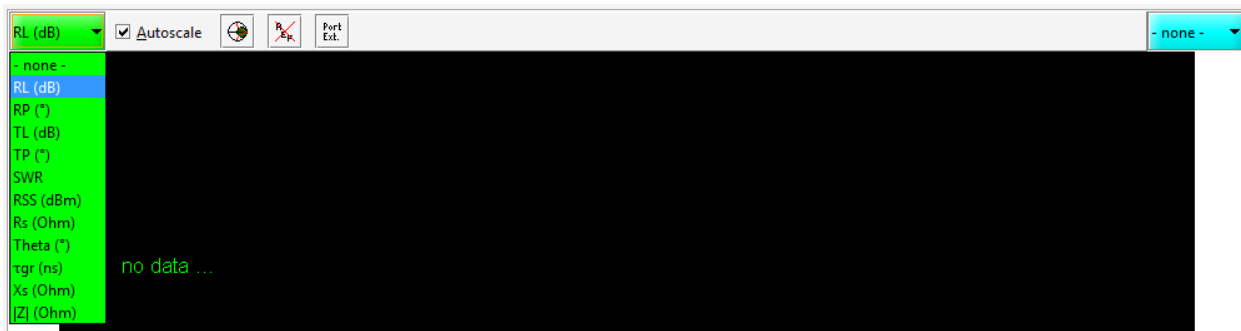
Delta: Calculates the absolute differences between most of the Marker 1 and Marker 2 data.

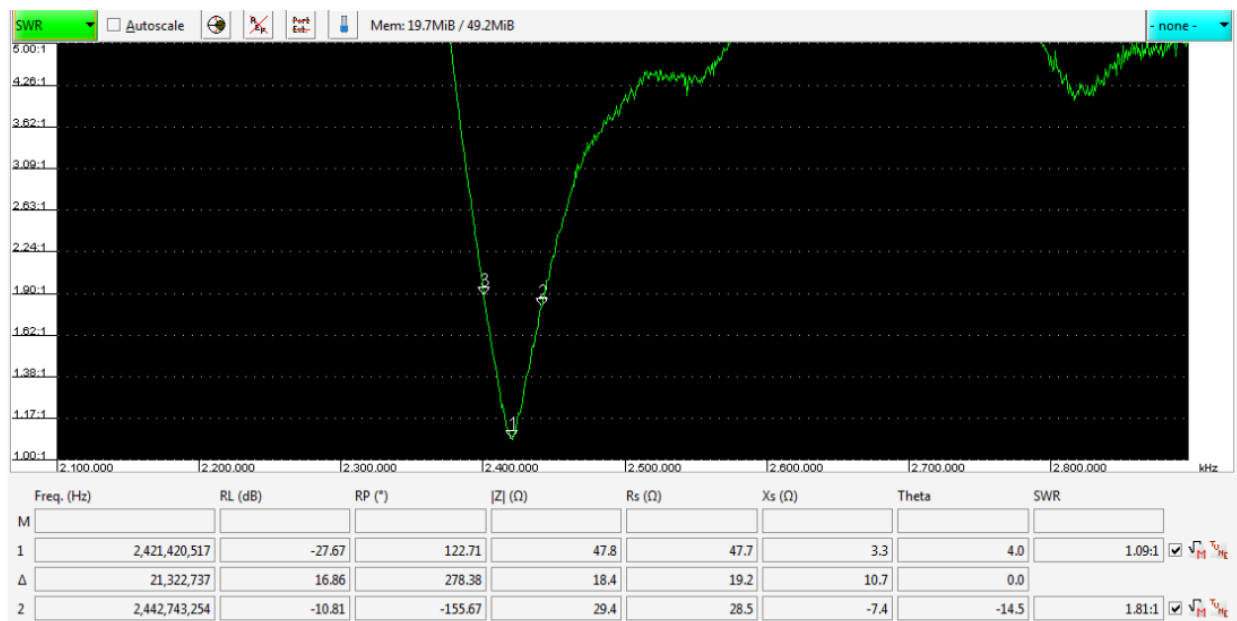
Marker 2: can be set by moving the mouse into the diagram panel and clicking the **left** mouse but-ton while pressing the **Shift** key. It can be moved using the mouse-wheel while pressing the **Shift** key.

Marker 3: can be set by moving the mouse into the diagram panel and clicking the **left** mouse but-ton while pressing the **Control**-key. It can be moved using the mouse-wheel while pressing the **Control**-key.

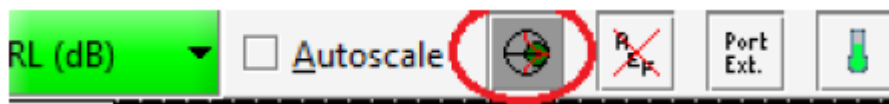
Marker 4: can be set by moving the mouse into the diagram panel and clicking the **left** mouse but-ton while pressing the **Shift-Control**-keys. It can be moved using the mouse-wheel while pressing the **Shift-Control**-keys.

- To observe other characteristics such as SWR, Phase, Impedance, etc., go to left green color drop down option and select the relevant option and right blue color drop down option select as none to remove TP graph.





8. Click Smithchart option to see magnitude & Phase of RMSA antenna



9. Go to Menu > export Save graph in PDF,S-parameter file JPG and CSV file.

Note: Now you can remove RMSA antenna and connect other antenna TMSA

Similarly, you can measured return loss of other antennas like CMSA or TMSA using same method.